

WEEK 7 – POWER BI REPORT
STUDENT PERFORMANCE ANALYSIS

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1. Introduction

The **Student Performance Analysis** project is developed using **Microsoft Power BI** to analyze and visualize students' academic performance. The dataset contains student details and marks obtained in **Math, Reading, and Writing**.

Power Query was used to clean and transform the raw dataset. Separate tables were created for **Students Marks, Students Performance, and Students Details**. Calculated columns and DAX measures were then used to analyze the students' overall performance.

2. Objectives

The main objectives of this project are:

- To clean and transform student data using Power Query.
- To analyze Math, Reading, and Writing scores.
- To calculate Total Marks and Average Score.
- To classify students based on Grade.
- To identify Pass and Fail students.
- To compare performance based on gender and other student details.
- To create an interactive Power BI dashboard.

3. Dataset Description

The dataset contains information about **1000 students**.

The major attributes include:

Column	Description
Student_ID	Unique identification number
Gender	Gender of the student
Race/Ethnicity	Student group
Parental Level of Education	Parent's education level
Lunch	Lunch category
Test Preparation Course	Preparation course status
Math Score	Mathematics marks
Reading Score	Reading marks
Writing Score	Writing marks

4. Data Cleaning Using Power Query

The raw dataset was imported into Power BI and opened in **Power Query Editor**.

The following transformations were performed:

- Promoted the first row as headers.
- Changed the required data types.
- Capitalized text values.

- Trimmed unnecessary spaces.
- Cleaned text data.
- Reordered columns.
- Added index columns.
- Removed unnecessary columns.
- Removed duplicate records.
- Converted marks into numeric data types.
- Rounded calculated values where required.

WEEK 7-Student Performance

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Whole Number Merge Queries Use First Row as Headers Append Queries Replace Values Combine Files Combine

Queries [4]

StudentsPerformance

Students Marks

Students details

Measures Table

Table.TransformColumnTypes(#"Rounded Off",{{"Total Marks", Int64.Type}})

Student_ID	gender	race/ethnicity	parental level of education	lunch
1	Female	Group B	Bachelor's Degree	Standard
2	Female	Group C	Some College	Standard
3	Female	Group B	Master's Degree	Standard
4	Male	Group A	Associate's Degree	Free/Reduced
5	Male	Group C	Some College	Standard
6	Female	Group B	Associate's Degree	Standard
7	Female	Group B	Some College	Standard
8	Male	Group B	Some College	Free/Reduced
9	Male	Group D	High School	Free/Reduced
10	Female	Group B	High School	Free/Reduced
11	Male	Group C	Associate's Degree	Standard
12	Male	Group D	Associate's Degree	Standard
13	Female	Group B	High School	Standard
14	Male	Group A	Some College	Standard
15	Female	Group A	Master's Degree	Standard
16	Female	Group C	Some High School	Standard
17	Male	Group C	High School	Standard
18	Female	Group B	Some High School	Free/Reduced
19	Male	Group C	Master's Degree	Free/Reduced
20	Female	Group C	Associate's Degree	Free/Reduced
21	Male	Group D	High School	Standard
22	Female	Group B	Some College	Free/Reduced

Query Settings

PROPERTIES

Name

StudentsPerformance

APPLIED STEPS

Source

Promoted Headers

Changed Type

Capitalized Each Word

Trimmed Text

Cleaned Text

Reordered Columns

Added Index

Removed Columns

Added Index1

Renamed Columns

Reordered Columns1

Added Custom

Removed Duplicates2

Added Custom1

Changed Type1

Added Conditional Column1

Added Conditional Column

Reordered Columns2

Removed Duplicates

5. Students Marks Table

A separate **Students Marks** table was created for subject-wise analysis.

The Math, Reading, and Writing score columns were transformed using **Unpivot Columns**.

The final table contains:

- Student_ID
- Subjects
- Score
- Example:

Student_ID	Subjects	Score
1	Writing Score	74
1	Reading Score	72
1	Math Score	72

This structure makes it easier to compare scores across different subjects.

Query: Table.RenameColumns(#"Unpivoted Columns",({"Attribute", "Subjects"}, {"Value", "Score"}))

Student_ID	Subjects	Score
1	writing score	74
1	reading score	72
1	math score	72
2	writing score	88
2	reading score	90
2	math score	69
3	writing score	93
3	reading score	95
3	math score	90
4	writing score	44
4	reading score	57
4	math score	47
5	writing score	75
5	reading score	78
5	math score	76
6	writing score	78
6	reading score	83
6	math score	71
7	writing score	92
7	reading score	95
7	math score	88
8	writing score	39

APPLIED STEPS:

- Cleaned Text
- Reordered Columns
- Added Index
- Removed Columns
- Added Index1
- Renamed Columns
- Reordered Columns1
- Added Custom
- Removed Duplicates2
- Added Custom1
- Changed Type1
- Added Conditional Column1
- Added Conditional Column
- Reordered Columns2
- Removed Duplicates
- Rounded Off
- Changed Type2
- Removed Other Columns
- Unpivoted Columns
- Renamed Columns1

6. Students Details Table

The **Students Details** table contains the demographic and background information of students.

The table includes:

- Student_ID
- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course
- Grade
- Result

Duplicate values were removed and the data was cleaned before creating relationships.

WEEK 7-Student Performance

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Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Columns Keep Remove Rows Split Column Group By Data Type: Whole Number Merge Queries Use First Row as Headers Append Queries Replace Values Combine Files

Queries [4]

StudentsPerformance Students Marks Students details Measures Table

fx = Table.RemoveColumns(#"changed Type2",{"writing score", "reading score", "math score"})

	Student_ID	gender	race/ethnicity	parental level of education	lunch
1	1	Female	Group B	Bachelor'S Degree	Standard
2	2	Female	Group C	Some College	Standard
3	3	Female	Group B	Master'S Degree	Standard
4	4	Male	Group A	Associate'S Degree	Free/Reduced
5	5	Male	Group C	Some College	Standard
6	6	Female	Group B	Associate'S Degree	Standard
7	7	Female	Group B	Some College	Standard
8	8	Male	Group B	Some College	Free/Reduced
9	9	Male	Group D	High School	Free/Reduced
10	10	Female	Group B	High School	Free/Reduced
11	11	Male	Group C	Associate'S Degree	Standard
12	12	Male	Group D	Associate'S Degree	Standard
13	13	Female	Group B	High School	Standard
14	14	Male	Group A	Some College	Standard
15	15	Female	Group A	Master'S Degree	Standard
16	16	Female	Group C	Some High School	Standard
17	17	Male	Group C	High School	Standard
18	18	Female	Group B	Some High School	Free/Reduced
19	19	Male	Group C	Master'S Degree	Free/Reduced
20	20	Female	Group C	Associate'S Degree	Free/Reduced
21	21	Male	Group D	High School	Standard
22	22	Female	Group B	Some College	Free/Reduced

Query Settings

PROPERTIES

Name: Students details

APPLIED STEPS

- Capitalized Each Word
- Trimmed Text
- Cleaned Text
- Reordered Columns
- Added Index
- Removed Columns
- Added Index1
- Renamed Columns
- Reordered Columns1
- Added Custom
- Removed Duplicates2
- Added Custom1
- Changed Type1
- Added Conditional Column1
- Added Conditional Column
- Reordered Columns2
- Removed Duplicates
- Rounded Off
- Changed Type2
- Removed Columns1

7. Students Performance Table

The **StudentsPerformance** table was created to combine the student information with their academic scores.

The table contains:

- Student_ID
- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Math Score
- Reading Score
- Writing Score
- Total Marks
- Average
- Grade
- Result

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Queries [4] X ✓ fx = Table.TransformColumnTypes(*"Rounded Off",{"Total Marks", Int64.Type}))

StudentsPerformance

	123 Student_ID	APC gender	APC race/ethnicity	APC parental level of education	APC lunch
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	1	Female	Group B	Bachelor's Degree	Standard
2	2	Female	Group C	Some College	Standard
3	3	Female	Group B	Master's Degree	Standard
4	4	Male	Group A	Associate's Degree	Free/Reduced
5	5	Male	Group C	Some College	Standard
6	6	Female	Group B	Associate's Degree	Standard
7	7	Female	Group B	Some College	Standard
8	8	Male	Group B	Some College	Free/Reduced
9	9	Male	Group D	High School	Free/Reduced
10	10	Female	Group B	High School	Free/Reduced
11	11	Male	Group C	Associate's Degree	Standard
12	12	Male	Group D	Associate's Degree	Standard
13	13	Female	Group B	High School	Standard
14	14	Male	Group A	Some College	Standard
15	15	Female	Group A	Master's Degree	Standard
16	16	Female	Group C	Some High School	Standard
17	17	Male	Group C	High School	Standard
18	18	Female	Group B	Some High School	Free/Reduced
19	19	Male	Group C	Master's Degree	Free/Reduced
20	20	Female	Group C	Associate's Degree	Free/Reduced
21	21	Male	Group D	High School	Standard
22	22	Female	Group B	Some College	Free/Reduced

Query Settings

PROPERTIES

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StudentsPerformance

APPLIED STEPS

Changed Type

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Queries [4] X ✓ fx = Table.TransformColumnTypes(*"Rounded Off",{"Total Marks", Int64.Type}))

StudentsPerformance

	APC test preparation course	123 writing score	123 reading score	123 math score	123 Total Marks
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	None	74	72	72	72
2	Completed	88	90	69	69
3	None	93	95	90	90
4	None	44	57	47	47
5	None	75	78	76	76
6	None	78	83	71	71
7	Completed	92	95	88	88
8	None	39	43	40	40
9	Completed	67	64	64	64
10	None	50	60	38	38
11	None	52	54	58	58
12	None	43	52	40	40
13	None	73	81	65	65
14	Completed	70	72	78	78
15	None	58	53	50	50
16	None	78	75	69	69
17	None	86	89	88	88
18	None	28	32	18	18
19	Completed	46	42	46	46
20	None	61	58	54	54
21	None	63	69	66	66
22	Completed	70	75	66	66

Query Settings

PROPERTIES

Name

StudentsPerformance

APPLIED STEPS

Changed Type

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Cleaned Text

Reordered Columns

Added Index

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Renamed Columns

Reordered Columns1

Added Custom

Removed Duplicates2

Added Custom1

Changed Type1

Added Conditional Column1

Added Conditional Column

Reordered Columns2

Removed Duplicates

Rounded Off

Changed Type2

Query Settings

Properties

Applied Steps

Changed Type

Capitalized Each Word

Trimmed Text

Cleaned Text

Reordered Columns

Added Index

Removed Columns

Added Index1

Renamed Columns

Reordered Columns1

Added Custom

Removed Duplicates2

Added Custom1

Changed Type1

Added Conditional Column1

Added Conditional Column

Reordered Columns2

Removed Duplicates

Rounded Off

Changed Type2

	math score	Total Marks	Average	Grade	Result
1	72	218	72.67	A	Pass
2	69	247	82.33	A	Pass
3	90	278	92.67	A	Pass
4	47	148	49.33	F	Pass
5	76	229	76.33	A	Pass
6	71	232	77.33	A	Pass
7	88	275	91.67	A	Pass
8	40	122	40.67	F	Pass
9	64	195	65	B	Pass
10	38	148	49.33	F	Pass
11	58	164	54.67	B	Pass
12	40	135	45	F	Pass
13	65	219	73	A	Pass
14	78	220	73.33	A	Pass
15	50	161	53.67	B	Pass
16	69	222	74	A	Pass
17	88	263	87.67	A	Pass
18	18	78	26	F	Fail
19	46	134	44.67	F	Pass
20	54	173	57.67	B	Pass
21	66	198	66	B	Pass
22	cc	710	70	B	Pass

8. Total Marks Calculation

The total marks were calculated by adding the three subject scores.

Formula:

Total Marks = Math Score + Reading Score + Writing Score

For example:

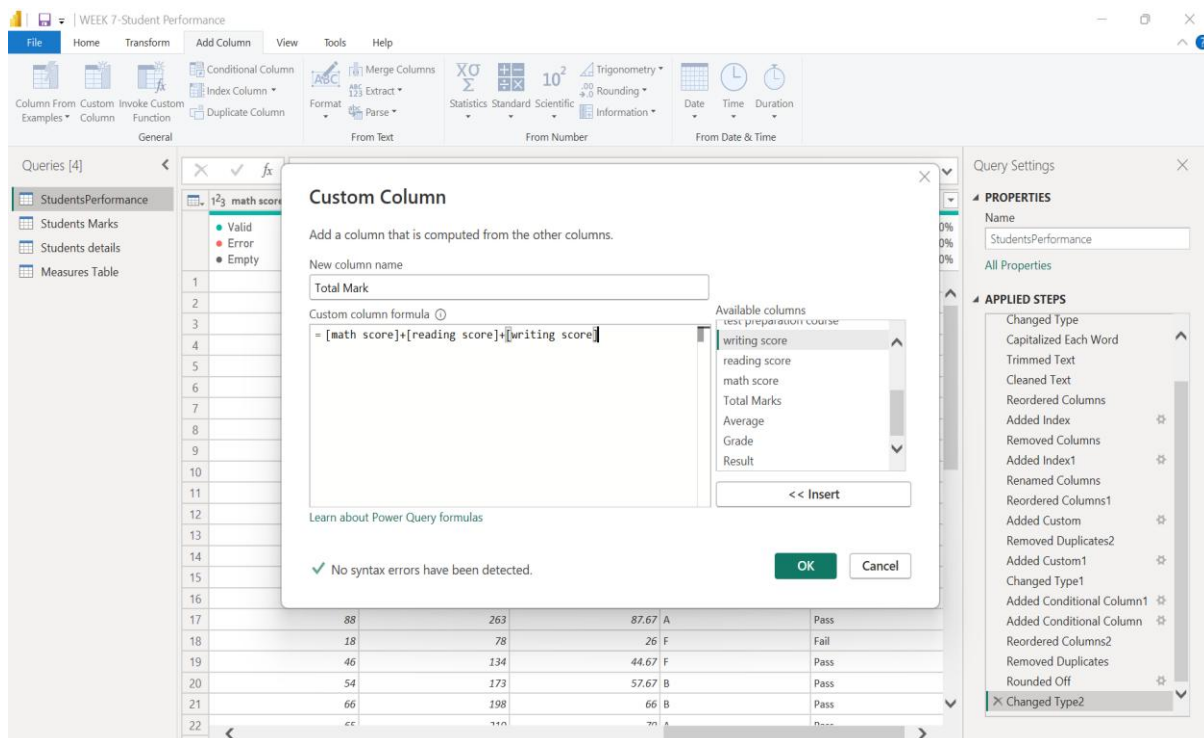
Math = 72

Reading = 72

Writing = 74

Total Marks = 218

This calculation helps to identify the student's overall marks.



9. Average Score Calculation

The **Average** column was created to calculate the average performance of each student.

Formula:

Average = Total Marks / 3

For example:

Total Marks = 218

Average = 218 / 3

= 72.67

The Average Score is used for grade classification and result analysis.

= Table.TransformColumnTypes("#Rounded Off",{{"Total Marks", Int64.Type}})									
	math score		123 Total Marks		1.2 Average		ABC 123 Grade		ABC 123 Result
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1			72		218		72.67	A	Pass
2			69		247		82.33	A	Pass
3			90		278		92.67	A	Pass
4			47		148		49.33	F	Pass
5			76		229		76.33	A	Pass
6			71		232		77.33	A	Pass
7			88		275		91.67	A	Pass
8			40		122		40.67	F	Pass
9			64		195		65	B	Pass
10			38		148		49.33	F	Pass
11			58		164		54.67	B	Pass
12			40		135		45	F	Pass
13			65		219		73	A	Pass
14			78		220		73.33	A	Pass
15			50		161		53.67	B	Pass
16			69		222		74	A	Pass
17			88		263		87.67	A	Pass
18			18		78		26	F	Fail
19			46		134		44.67	F	Pass
20			54		173		57.67	B	Pass
21			66		198		66	B	Pass
22			55		210		70	A	Pass

10. Grade Classification

A conditional column named **Grade** was created based on the Average Score.

The grades were classified using Power Query conditional logic.

Average Score Grade

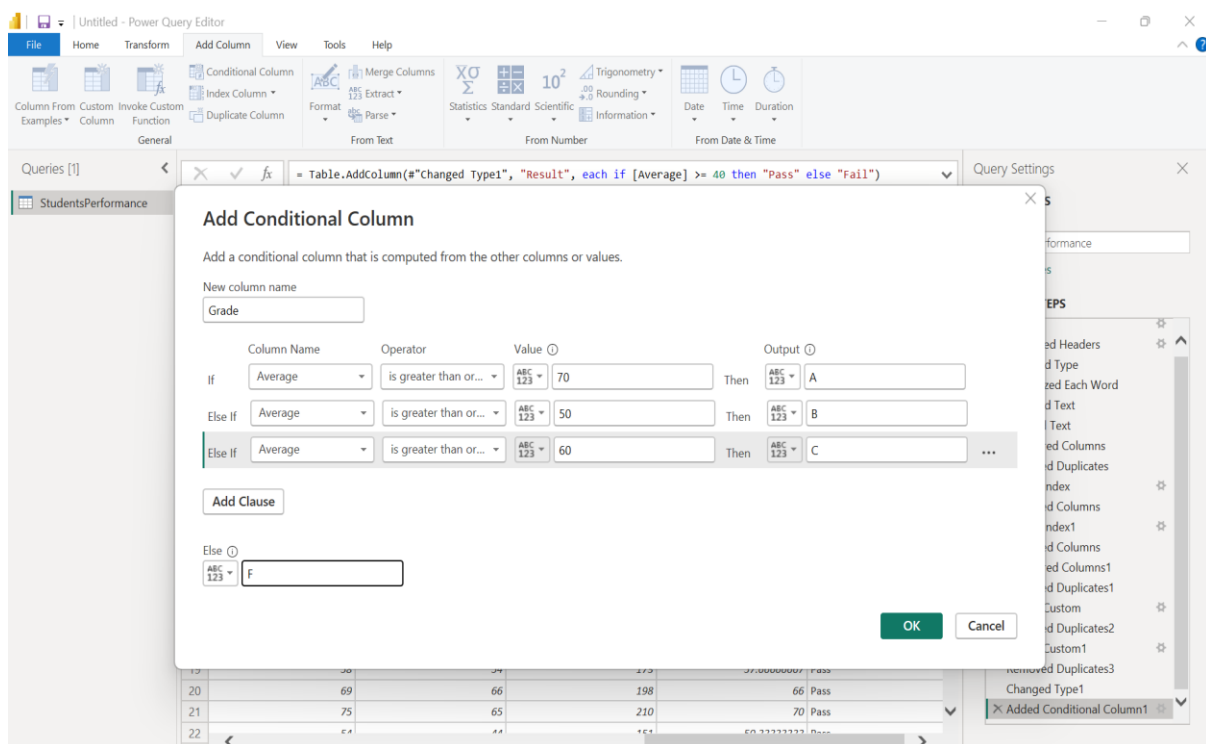
Above 70 A

Above 50 B

Above 60 C

Otherwise F

Note: In the current Power Query setup, the >50 condition appears before >60, so the >60 condition will not be reached for values between 60 and 70. If your intended classification is A >70, B >60, C >50, F otherwise, keep the conditions in descending order.



11. Result Classification

A **Result** column was created to classify students as Pass or Fail.

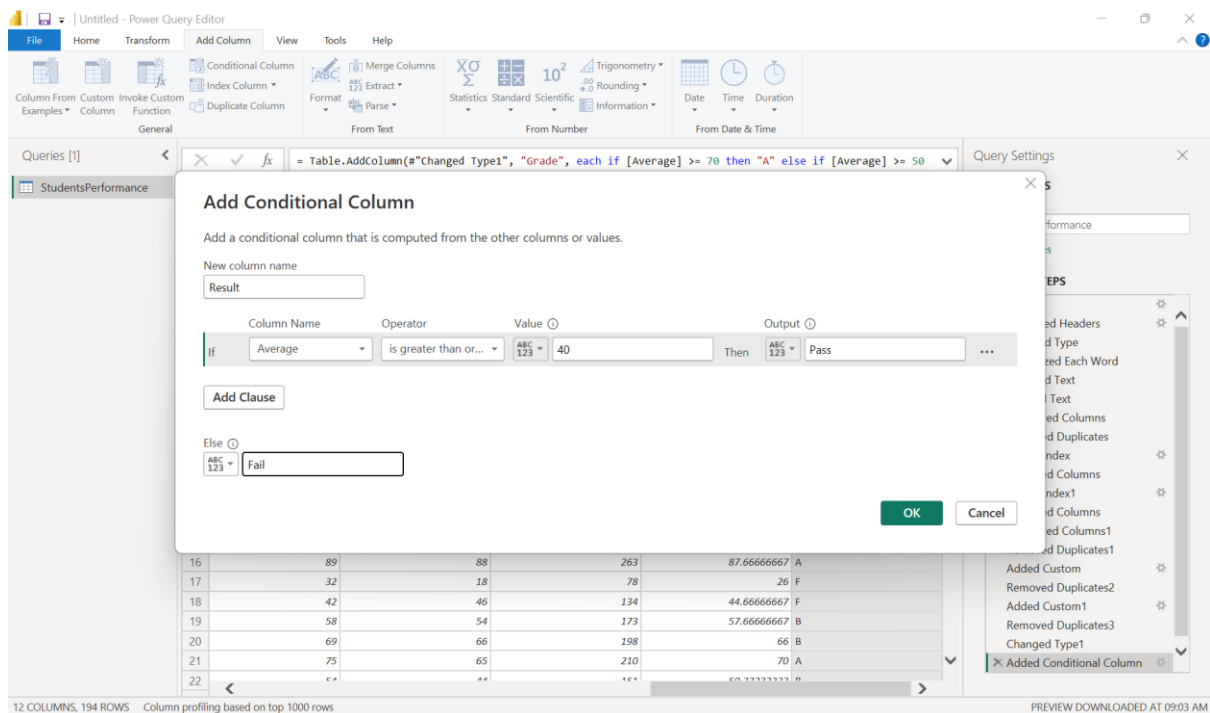
The condition used was:

If Average $\geq 40 \rightarrow$ Pass

Otherwise → Fail

Therefore:

- $\text{Average} \geq 40 \rightarrow \text{Pass}$
- $\text{Average} < 40 \rightarrow \text{Fail}$



12. Data Model

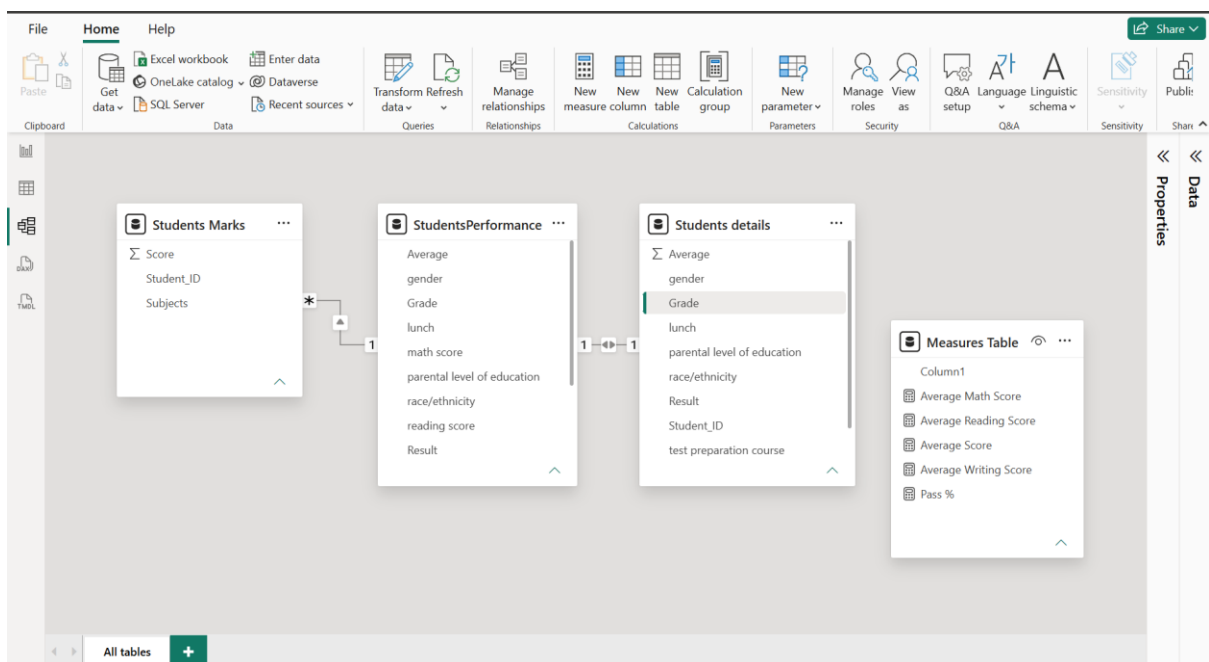
The Power BI model contains the following tables:

1. Students Marks
2. Students Performance
3. Students Details
4. Measures Table

The **Measures Table** contains the DAX measures required for dashboard analysis.

The main measures created are:

- Average Math Score
- Average Reading Score
- Average Score
- Average Writing Score
- Pass %



13. DAX Measures

Average Math Score

Average Math Score =

AVERAGE(StudentsPerformance[math score])

Average Reading Score

Average Reading Score =

AVERAGE(StudentsPerformance[reading score])

Average Writing Score

Average Writing Score =

AVERAGE(StudentsPerformance[writing score])

Average Score

Average Score =

AVERAGE(StudentsPerformance[Average])

Pass Percentage

Pass % =

DIVIDE(

 CALCULATE(

 COUNTROWS(StudentsPerformance),

 StudentsPerformance[Result] = "Pass"

),

 COUNTROWS(StudentsPerformance),

 0

)

The Pass % measure is formatted as a percentage in Power BI.

Conclusion

- Successfully imported and cleaned the student performance dataset using **Power BI**.
- Used **Power Query** for data cleaning and transformation.
- Created separate **Students Marks, Students Details, and Students Performance** tables.
- Used **Unpivot** to organize subject-wise marks.
- Calculated **Total Marks** and **Average Score**.
- Created **Grade** and **Result** classifications.
- Created **DAX measures** for subject-wise average scores and Pass Percentage.
- Established relationships between the required tables.
- The project helped in understanding **data cleaning, transformation, calculation, and analysis using Power BI**.